



Video-based adaptive respiratory rate monitoring for clinical applications

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Abstract

Respiratory rate (RR) is a critical physiological parameter for assessing respiratory function and is widely used in clinical applications. Non-contact monitoring of the RR enables continuous monitoring, avoiding the discomfort often associated with traditional contact methods, making it highly desirable in clinical applications. The existing method [1] could measure the RR from RGB videos by combining the optical flow method and the negative feedback crossover point method. The selection of feature points in the optical flow method significantly affects the accuracy of respiratory signals. To enhance the dynamic tracking performance of feature points, we propose to simplify the feature point evaluation criteria. Additionally, we utilize the temporal information of the feature points to adaptively optimize the feature point space. The crossover point method is prone to erroneous crossover points, while the negative feedback crossover point method can remove erroneous points but may also eliminate valid ones. Therefore, we further optimize the crossover point removal strategy based on the physiological information of the crossover points. In the Large-scale Bedside Respiration Dataset for Intensive Care (LBRD-IC), our proposal method estimates the RR with a mean absolute error (MAE) of 4.034 beats per minute (BPM) and a mean squared error (MSE) of 30.34 BPM², while the existing method achieves a MAE of 4.464 BPM and a MSE of 36.75 BPM². This demonstrates the effectiveness of our proposal method in clinical applications. The codes are publicly available at <https://github.com/ZinChou/RR>.

Keywords Respiratory rate · Clinical applications · Non-contact monitoring · Optical flow

1 Introduction

Respiration is a crucial physiological activity for maintaining the normal metabolism and physiological functions of cells in the human body. Research has shown that the respiratory rate (RR) can serve as a predictive indicator for cardiac arrest [2, 3]; abnormal respiratory patterns may signal sleep disorders, such as sleep apnea [4]; moreover, by analyzing respiratory signals, one can infer a person's emotional state [5, 6]; irregular variations in the RR are often among

the early symptoms of various diseases, such as respiratory infections, pneumonia, heart failure, and sepsis [7]. In intensive care units (ICU) or post-surgical recovery, the RR is a key indicator of patient stability and recovery. Therefore, monitoring patients' RR in clinical settings is essential for medical care.

Currently, RR monitoring methods are mainly divided into two categories: contact and non-contact methods. Traditionally, this monitoring relies on contact devices, which measure the RR using sensors attached to the patient's body. However, patient movements can affect the position and stability of the sensors, leading to measurement interruptions and a lack of continuity in the data [8]. Furthermore, the use of contact sensors may interfere with nurses' daily care activities, such as cleaning patients and changing their dressings. This not only reduces the efficiency of nursing workflows but also affects the comfort of the patient [9, 10]. The non-contact methods effectively address these issues. Among these, RGB video-based methods have gained widespread attention due to their lightweight and accessible

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hardware [11–14]. The optical flow method in tandem with crossover point method is a commonly used algorithm for extracting the RR from RGB videos [1].

The optical flow method is a well-established technique widely used for object motion tracking and prediction [15]. The optical flow can be utilized to recover respiratory motion from videos captured by RGB cameras. In traditional optical flow methods, there are many redundant feature points, which not only increase the time consumption of the algorithm but also decrease the accuracy of respiratory signals. To address these issues, Tan et al. [1] proposed a novel optical flow method. They employed a feature point selection strategy based on Harris corner detection, which aggregated spatial information to choose feature points rich in respiratory information. This significantly reduced the algorithm's time consumption while improving accuracy. However, they overlooked the dynamic tracking performance of feature points, which limited the potential for further accuracy enhancement. Additionally, after extracting the respiratory signals from feature points, they employed a correlation-guided method that aggregated temporal information and discarded feature points with weak correlation to respiratory motion using a static threshold, thereby enhancing the accuracy of the respiratory signal. But this method failed to account for environmental conditions and individual differences among subjects, resulting in unstable performance.

After the respiratory signal is obtained, the crossover point method is commonly used to extract the RR from respiratory signals. It calculates RR by counting the crossover points between the respiratory waveform and its time-shifted version. Although this method is simple to implement, it is prone to errors caused by erroneous crossover points, leading to issues with estimation accuracy. To address these shortcomings, Tan et al. [1] developed a variant called the negative feedback crossover point method, which iteratively eliminated erroneous crossover points based on the obtained RR, thereby improving the accuracy of the RR estimation. However, the crossover point elimination method in the negative feedback method has some issues, which may result in the removal of valid crossover points and a decrease in the accuracy of the RR extraction.

To address the issues present in the current RR extraction algorithm based on optical flow and crossover point methods, we propose an adaptive RR extraction algorithm. Our specific contributions are as follows:

- To enhance the dynamic tracking performance of the feature points, we simplify the feature point evaluation criteria by focusing on the minimum eigenvalue of the feature matrix in our feature point selection strategy.
- We propose calculating the threshold based on the correlation coefficients between the magnitude vectors of

the feature points and the respiration signal, and then removing feature points that do not meet the threshold. This method aims to improve the accuracy and robustness of the algorithm across varying monitoring environments and subjects.

- We propose using physiological information of crossover points to guide the identification and removal of erroneous crossover points, ensuring that erroneous crossover points are removed while avoiding the removal of correct ones.

The rest of the paper is organized as follows: Section 2 reviews related work on non-contact RR monitoring. Section 3 details our improved RR extraction algorithm. Section 4 introduces the dataset and experiments on our improved algorithm. Finally, the conclusions are drawn in Sect. 5.

2 Related work

In recent years, non-contact RR extraction methods have attracted increasing attention across various application fields due to their significant theoretical value and broad application prospects [16]. These non-contact RR extraction methods can be primarily classified into four categories [17]: 1) methods based on respiratory sounds; 2) methods based on temperature; 3) methods based on heart activity modulation; 4) methods based on chest wall motion.

2.1 Based on respiratory sounds

These methods involve processing and analyzing recorded respiratory audio to extract respiratory signals. For example, Nam et al. [18] proposed utilizing Welch's periodogram and autoregressive spectrum methods to perform spectral analysis on nasal respiratory recordings made with a smartphone in order to obtain the RR. This method can still accurately estimate the RR even when the smartphone microphone is placed 30 cm away from the nostrils. Kumar et al. [19] explored a model-based RR estimation approach that extracted relevant information from short audio segments recorded from healthy adults after exertion. The study implemented a multi-task long short-term memory (LSTM) network [20] with convolutional layers to process Mel-filterbank energy features, enabling accurate RR estimation under varying background noise conditions. While these methods have demonstrated high effectiveness both theoretically and experimentally, their performance in real-world applications is often strongly influenced by environmental sound conditions. Environmental noise, such as conversation, traffic, or other background sounds, can interfere with the respiratory sound waves captured by the microphone,

thereby affecting the accuracy of the spectral analysis and leading to increasing estimation errors in the RR.

2.2 Based on temperature

Such methods detect the thermal changes caused by human respiration to extract the RR. Barbosa et al. [21] proposed an algorithm for monitoring RR using thermal imaging, estimating the respiratory frequency through the temperature fluctuations below the nose during the respiratory cycle. Hu et al. [22] developed a nighttime RR measurement system based on infrared imaging. It combines a custom cascade face classifier to detect the facial region in infrared images. Through discriminative regression and coordinate transformation, the system identifies the region of interest (ROI), including the mouth and nose. A spatiotemporal context learning tracking algorithm is then employed to extract the RR signal, followed by signal processing using time-domain analysis methods. Farah et al. [23] investigated the use of thermal imaging cameras to monitor RR in children. The study compared the RR measurements obtained using the thermal imaging camera with those obtained from standard methods, such as nasal thermistor, respiratory impedance plethysmography, and transcutaneous CO₂ measurement. The results demonstrated a strong correlation ($r^2 = 0.994$) between thermal imaging and the standard methods.

The advantage of these methods lies in their relatively low susceptibility to interference from movements in other parts of the body, enabling stable monitoring under various conditions. This makes them particularly suitable for monitoring the RR during physical activity. However, the technology also has certain limitations, such as its high sensitivity to environmental temperature changes. Fluctuations in ambient temperature or local temperature differences can impact the measurement results, leading to inaccuracies in the data.

2.3 Based on heart activity modulation

These methods are based on the close physiological relationship between various cardiac activities and respiratory motion, using signals such as electrocardiograms (ECG) and photoplethysmograms (PPG) to measure respiratory signals [24, 25]. Mazzanti et al. [26] proposed a method that automatically selects the optimal lead from a multi-lead system, calculates the electrocardiogram-derived respiration (EDR) based on changes in the QRS complex of the ECG, and derives the respiratory waveform through interpolation. In the time domain, the RR is calculated, with the respiratory cycle identified using a min-max detection method. Zhao et al. [27] proposed a signal processing technique for extracting the EDR from a single-lead ECG. The

method utilizes wavelet transform to process the ECG signal, followed by filtering to extract the EDR. A comparative analysis with conventional respiratory measurements demonstrated a strong similarity between the extracted EDR and actual respiratory signals.

Among various signals, remote-PPG (rPPG) [22, 28] is most commonly used, as it contains multidimensional respiratory information compared to other signals. This method estimates the RR by analyzing the small changes in skin color caused by variations in blood oxygen levels induced by respiratory motion. Chen et al. [29] proposed a non-contact video-based physiological measurement method, utilizing a new motion representation based on a skin reflection model and a new attention mechanism that guides motion estimation using appearance information. These two mechanisms provide robust measurements under heterogeneous lighting and significant motion. Osathitporn et al. [30] proposed a continuous estimation method for RR, called RRWaveNet. RRWaveNet is a compact, end-to-end deep learning model that requires no feature engineering and can use low-cost raw PPG signals as input.

2.4 Based on chest wall motion

These methods are typically categorized into marker-based motion capture methods [31, 32], depth camera methods [33, 34], laser vibration measurement methods [35], radio frequency sensor methods [36], and RGB camera-based methods [15, 37, 38], all of which fundamentally extract respiratory signals by capturing the motion of the chest and abdomen. Typically, the motion of the chest and abdomen is more pronounced and observable than the skin color changes induced by cardiac regulation, making it more suitable for the accurate extraction of respiratory signals. Wang et al. [34] utilized a RGB camera in collaboration with marker tracking as a data acquisition sensor to achieve low-cost and remote measurement of respiratory signals. Kempfle et al. [33] used a depth camera to capture the slight distance changes in the user's chest over time to remotely detect an individual's respiration, and proposed a new method to simulate chest elevation for robustly monitoring the user's respiration while seated or standing and facing the camera. Shyu et al. [36] proposed a new multi-feature alignment (MFA) two-layer EEMD method to address signal interference issues in dynamic human detection with ultra-wideband (UWB) radar. This method is capable of simultaneously detecting slight respiration and heartbeat information from a moving driver. Tran et al. [15] proposed a low-cost system that uses real-time images to detect human respiration and estimate the peak of inhalation to trigger X-ray imaging. This method applies

the Lucas-Kanade algorithm to extract chest features and adjusts the pyramid Lucas-Kanade to track small motion.

3 Methodology

3.1 Overview

As shown in Fig. 1, the algorithm of extracting RR from RGB videos is divided into two main parts: respiratory signal extraction and RR calculation. Using the feature point selection strategy, the positions of the feature points tracked by the optical flow method in the first frame are selected and represented as $\mathbf{p}(i, 1) = [p_x(i, 1), p_y(i, 1)]^T$, where $i = \{1, 2, \dots, K\}$ and K denotes the total number of feature points. Following [39], dense optical flow method is used to track the positions of the K feature points in each frame, denoted as $\mathbf{p}(i, j)$, where $j = \{1, 2, \dots, N\}$ and N represents the number of video frames. Based on the reference [40], the magnitude vector $\mathbf{A}(i) = [A(i, 1), A(i, 2), \dots, A(i, N)]^T$ for the i -th feature point can be calculated using Eq. (1). Using Eq. (2), the magnitude vectors of all feature points can be combined to form the respiratory signal $\mathbf{Resp}$.

$$A(i, j) = \sqrt{p_x(i, j)^2 + p_y(i, j)^2}, \quad (1)$$

$$\mathbf{Resp}(j) = \frac{1}{K} \sum_{i=1}^K A(i, j). \quad (2)$$

To enhance the accuracy of respiratory signal, we employ the adaptive correlation-guided method to eliminate feature points whose magnitude vectors are unrelated to respiratory motion, and subsequently, the respiratory signal $\mathbf{Resp}^*$ is recomputed using Eq. (2). The respiratory signal $\mathbf{Resp}^*$ is then processed through bandpass filtering and normalization

to obtain the respiratory signal $\mathbf{Resp}$. Finally, the RR is calculated using the negative feedback crossover point method based on $\mathbf{Resp}$.

3.2 Dynamic tracking-enhanced feature point selection strategy

In respiratory signal extraction, the selection of feature points tracked by optical flow methods is critical. The chosen feature points not only need to be easily trackable by the optical flow algorithm but must also contain significant respiratory information. To enhance the dynamic tracking performance of feature points, we introduce a feature point selection strategy based on the Shi-Tomasi corner detection [41]. Compared to the Harris corner detection [42], the Shi-Tomasi corner detection simplifies the evaluation criterion by focusing on the minimum eigenvalue of the feature matrix, making corner detection more accurate in dynamic environments, and thus more suitable for feature point tracking in such conditions. Additionally, to ensure that the feature points contain high-quality respiratory motion information, we employ spatial information of the feature points for selection as reference in [1].

To detect the feature points, feature matrix $\mathbf{M}$ is utilized to reflect the texture and variations of local regions centered at (p_x, p_y) and is defined as follows:

$$\mathbf{M} = \sum_{x,y} \omega(x, y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}, \quad (3)$$

where ω represents the window function, typically a Gaussian weight used for smoothing, while I_x and I_y represent the gradients at the point (x, y) along the X and Y axes, respectively.

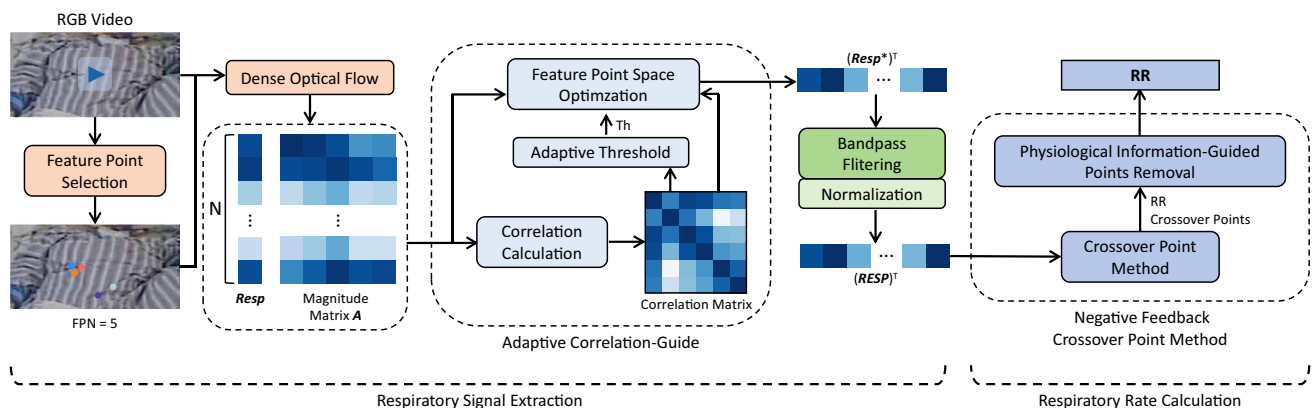


Fig. 1 Flowchart for extracting the RR from clinically captured RGB videos. In the flowchart, the respiratory signal is represented as a vector, where FPN denotes the number of feature points, $\mathbf{Resp}$ denotes

the respiratory signal, $\mathbf{Th}$ denotes the threshold, and $(\cdot)^T$ denotes the transposition operation

To more intuitively display the texture features of feature points, we perform eigenvalue decomposition on the feature matrix M :

$$M = H \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} H^{-1}, \quad (4)$$

where matrix H is composed of the eigenvectors of the feature matrix M , which are used for eigenvalue decomposition, and both λ_1 and λ_2 are the eigenvalues of the feature matrix. λ_1 and λ_2 reflect the rates of change in grayscale intensity in the directions of the X and Y axes, respectively. Based on λ_1 and λ_2 , the corner function can be defined as follows:

$$R = \min(\lambda_1, \lambda_2). \quad (5)$$

R only reflects texture features and tracking performance of the points. To ensure that the evaluation function considers not only the tracking performance of the feature points but also the respiratory motion information, a distance function D is introduced to aggregate spatial information.

Assuming each frame contains N pixels, and based on this assumption, the center coordinates are defined as:

$$\begin{cases} \bar{s}_x = \frac{1}{N} \sum_{i=1}^N s_{x,i}, \\ \bar{s}_y = \frac{1}{N} \sum_{i=1}^N s_{y,i}. \end{cases} \quad (6)$$

According to Eq. (6), the definition of the distance function is as follows:

$$D = \exp \left(-\frac{(p_x - \bar{s}_x)^2 + (p_y - \bar{s}_y)^2}{2\sigma^2} \right), \quad (7)$$

where D represents the Gaussian weight based on the Euclidean distance between the point (p_x, p_y) and the center $(\bar{s}_x, \bar{s}_y)$, and σ is the standard deviation that controls the spread of the weight function, and is typically defined as $\sigma = \frac{w+h}{4}$, where w and h are the width and height of the image, respectively.

The corner response function R captures texture features and tracking performance of the points, while the distance function D contains the respiratory motion information. By combining these two functions, the image information can be fully utilized, ensuring that the feature points not only have high tracking performance but also contain significant respiratory motion information. Based on Eq. (5) and Eq. (7), the feature point quality function Q is defined as:

$$Q = R \cdot D. \quad (8)$$

As shown in Fig. 2 (b), when Q is greater than Q_{th} , the point is selected as the feature point tracked by the optical flow method. Q_{th} is defined as follows: Suppose the values of the quality function Q for all points are sorted in descending order as $Q = [Q_1, Q_2, \dots, Q_{P_N}]$, where P_N represents the total number of points. Then, $Q_{th} = Q_K$, where K denotes the number of feature points.

3.3 Adaptive correlation-guided method

The feature point selection strategy utilizes the spatial information from the first frame of the RGB video to filter feature

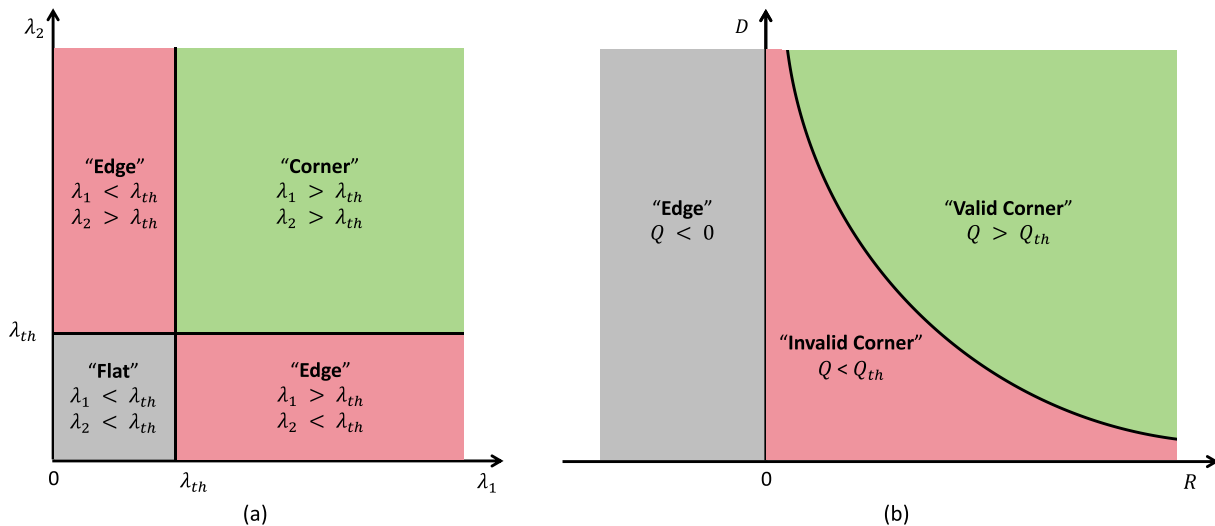


Fig. 2 Diagram of feature point selection method: **a** Diagram of Shi-Tomasi corner selection strategy; **b** Diagram of feature point selection strategy combining distance information with Shi-Tomasi corner features. λ_1 , λ_2 represent the eigenvalues of the feature matrix M ,

and λ_{th} is the corner judgment threshold. R is the Shi-Tomasi corner response function, D is the distance function, and $Q = R \cdot D$ is the feature point quality function

points for optical flow tracking. However, the selected feature points may still include those with weak correlations to respiratory motion. Additionally, during the optical flow tracking process, noise interference may impact the accuracy of the feature points' magnitude vectors, which contain the temporal information. To address this, a straightforward method is to set a static correlation threshold to exclude feature points whose magnitude vectors have weak correlations with the respiratory signal. However, this is not ideal, as the overall correlation between the magnitude vectors of feature points and respiratory signal can vary depending on the subjects and noise conditions. When the overall correlations are high, a higher threshold can help remove feature points whose magnitude vectors are relatively more influenced by noise, thereby improving the accuracy of

the respiratory signal. As shown in Fig. 3, the magnitude vector corresponding to the second feature point exhibits a good correlation with the respiratory signal. However, it is relatively more affected by noise, and retaining it would compromise the accuracy of the extracted respiratory signal. Conversely, when the overall correlations are relatively low, a lower threshold can retain more respiratory information while reducing extreme situations, thus making the method more stable. The static threshold fails to effectively adapt to these varying conditions.

To effectively address the above issues and improve both the accuracy and robustness of the respiratory signal, we propose an adaptive feature point space optimization method based on the temporal information of feature points. This method can more flexibly adapt to varying signal conditions and noise levels.

Based on the magnitude vector $A(i)$ of the i -th feature point, $i = \{1, \dots, K\}$, a magnitude matrix $A = [A(1), A(2), \dots, A(K)]$ for the feature points can be constructed. The respiratory signal $Resp$ can be obtained using Eq. (2). Thus, an augmented matrix $RA = [Resp, A]$ containing both the respiratory signal and the feature point magnitude matrix can be obtained. For convenience in subsequent discussions, an alternative representation of the augmented matrix is defined as follows:

$$RA = [RA_0, RA_1, \dots, RA_{K-1}, RA_K]. \quad (9)$$

The covariance matrix of the augmented matrix RA is represented as follows:

$$C = (c_{ij})_{K \times K} = \begin{bmatrix} c_{00} & c_{01} & \dots & c_{0K} \\ c_{10} & c_{11} & \dots & c_{1K} \\ \vdots & \vdots & \ddots & \vdots \\ c_{K0} & c_{K1} & \dots & c_{KK} \end{bmatrix}. \quad (10)$$

In the correlation matrix, the value c_{0k} , where $k = \{1, 2, \dots, K\}$, represents the Pearson correlation coefficient [43] between the respiratory signal $Resp$ and the magnitude vector of the k -th feature point. A smaller value indicates that the magnitude vector of the k -th feature point has a weaker correlation with the respiratory signal $Resp$.

To quantify the overall correlation between the magnitude vectors and respiratory motion, as well as the variability in correlations across different magnitude vectors, we computed the mean and standard deviation of the correlation coefficients for all feature points:

$$c_{mean} = \frac{1}{K} \sum_{k=1}^K c_{0k}, \quad (11)$$

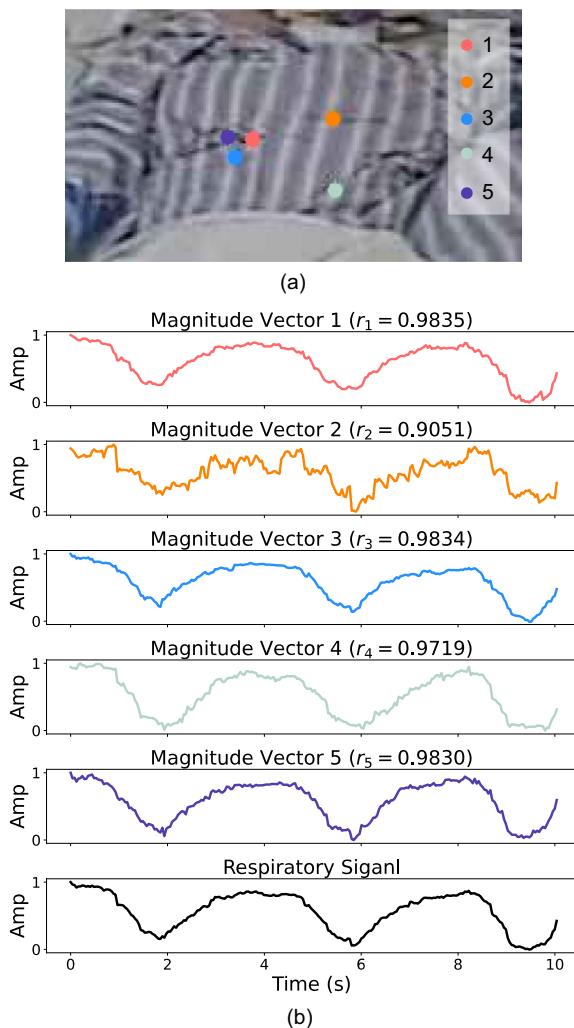


Fig. 3 **a** Five feature points selected by the improved feature point selection strategy. **b** Respiratory signal and the magnitude vectors corresponding to the five feature points. The respiratory signal is the mean value of the five magnitude vectors. Amp represents signal amplitude. r_i denotes the Pearson correlation coefficient between the i -th magnitude vector and the respiratory signal

$$c_{std} = \sqrt{\frac{1}{K} \sum_{k=1}^K (c_{0k} - c_{mean})^2}, \quad (12)$$

where K represents the number of feature points, and c_{0k} represents the correlation coefficient of the k -th magnitude vector with the respiratory signal **Resp**.

Based on Eq. (11) and Eq. (12), the threshold for the correlation coefficient can be calculated using the following formula:

$$Th = \frac{c_{mean}}{\alpha^{c_{std}}}, \quad (13)$$

where α is a constant.

If $c_{0k} < Th$, where $k = \{1, 2, \dots, K\}$, the k -th feature point is removed. Based on the magnitude vectors of the remaining feature points, a new magnitude matrix $A^* = [A_1^*, A_2^*, \dots, A_{K-e}^*]$ is obtained, where e represents the number of removed feature points. Based on the new magnitude matrix A^* , the respiratory signal **Resp**^{*} can be computed using Eq. (2). The improved method not only enhances the accuracy of respiratory signal but also increases the robustness and reliability of the algorithm in dealing with variable subjects and noisy environments.

3.4 Physiological information-guided negative feedback crossover point method

The crossover point method is commonly used to extract the RR from respiratory signals, but its accuracy is easily affected by erroneous crossover points. The negative feedback crossover point method improves the accuracy of the RR estimation by removing erroneous crossover points. However, the negative feedback crossover point method may also remove valid crossover points. As shown in Fig. 4, the interval between crossover points B and C does

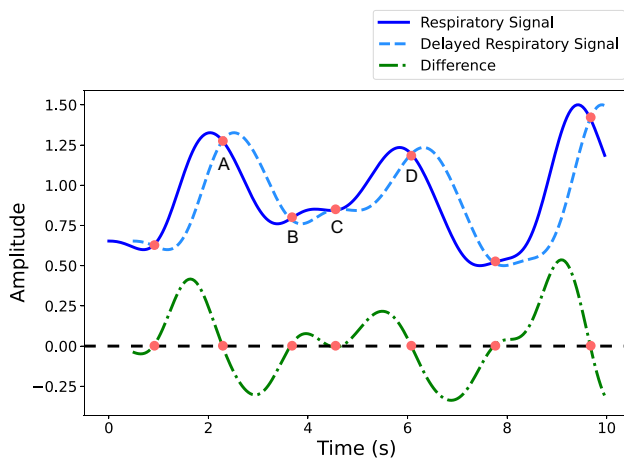


Fig. 4 Diagram of the crossover point method

not meet the minimum interval, so both points B and C are removed based on the negative feedback crossover point method. However, removing these points will result in the newly formed adjacent crossover points A and D having an excessively large distance. Moreover, in respiratory signals, peaks correspond to the local maxima, marking the end of inhalation, while troughs represent the local minima, indicating the end of exhalation. Both points A and D represent the transition from inhalation to exhalation, with a crossover point between them marking the transition from exhalation to inhalation. The removal of these points leads to a lower number of crossover points than the actual value, which consequently causes an underestimation of the RR. To address this issue, we propose a physiological information-guided negative feedback crossover point method.

Let the positions of all crossover points from the crossover point method be denoted as $P = [p_1, p_2, \dots, p_M]$, where M represents the total number of crossover points. The interval between k -th and $(k+1)$ -th crossover points is denoted as $\Delta p_k = p_{k+1} - p_k$, where $k = \{1, 2, \dots, M-1\}$. Based on the number of crossover points M , we can obtain the RR:

$$RR = \frac{M/2}{N/f_s} * 60, \quad (14)$$

where N is the total number of frames in the video sequence, and f_s is the frame rate of the video sequence.

Based on the RR , we can obtain the minimum adjacent crossover point interval $N_{minspan}$ and the maximum adjacent crossover point interval $N_{maxspan}$:

$$\begin{cases} N_{minspan} = \frac{30 * f_s * q_1}{RR}, \\ N_{maxspan} = \frac{30 * f_s * q_2}{RR}, \end{cases} \quad (15)$$

where q_1 and q_2 typically set to 0.6 and 1.4 based on experimental experience, respectively.

For each pair of adjacent crossover points p_k and p_{k+1} , if Δp_k is smaller than $N_{minspan}$, the crossover point p_{k+1} is first removed. If the crossover points p_{k-1} and p_{k+2} represent the same transition from inhalation to exhalation or from exhalation to inhalation, the crossover point p_k is regarded as a valid crossover point and retained. Otherwise, the distance d between the crossover points p_{k-1} and p_{k+2} is calculated:

$$d = \Delta p_k + \Delta p_{k+1} + \Delta p_{k+2}. \quad (16)$$

If d exceeds $N_{maxspan}$, the crossover point p_k is retained; otherwise, it is removed. This method ensures that the distribution of crossover points better aligns with the physiological meaning represented by the crossover points in the respiratory signal.

After comparing all adjacent crossover points, the remaining crossover point positions $P' = [p'_1, p'_2, \dots, p'_{M'}]$ are obtained, where M' is the number of remaining crossover points. Based on the number of remaining crossover points, the RR , $N_{\minspan}$ and $N_{\maxspan}$ can be recalculated to remove erroneous crossover points. This process is repeated iteratively until no crossover points are removed in a given iteration. The RR obtained in the last iteration is considered the final RR , denoted as RR^* .

This improvement ensures that valid crossover points are not mistakenly removed and that only erroneous crossover points caused by noise are removed. As a result, the accuracy of the algorithm in extracting RR are significantly enhanced.

3.5 Complexity analysis

To evaluate the computational efficiency of our improvements, we compare the theoretical time complexity of each module against that of the existing method.

- Feature Point Selection:** Both the existing and the proposed method compute a feature point quality function for each pixel, which consists of two components: a corner response function and a spatial distance function. In the existing method, the corner response is computed based on Harris corner detection, which involves calculating image gradients and constructing a local structure matrix $M \in \mathbb{R}^{2 \times 2}$ over a $w \times w$ window. This results in a per-pixel complexity of $\mathcal{O}(w^2)$. In our method, the Shi-Tomasi detector is used instead. While it introduces an additional eigenvalue decomposition step for M , this operation has constant-time complexity and is negligible compared to the construction of M . Therefore, the overall per-pixel complexity remains $\mathcal{O}(w^2)$. The spatial distance function in both methods is computed in constant time $\mathcal{O}(1)$. As a result, the total complexity for evaluating the feature point quality function across all N pixels is $\mathcal{O}(Nw^2)$. Subsequently, the top- K feature points are selected based on their quality scores. This selection step has a complexity of $\mathcal{O}(N \log K)$ using heap-based selection. Hence, the overall time complexity of the feature point selection module for both the existing and the proposed method is: $\mathcal{O}(Nw^2 + N \log K)$.
- Correlation-Guided Filtering:** Both the existing method and our proposed adaptive strategy compute the pairwise Pearson correlation coefficients between the magnitude vectors of all K feature points. This results in a correlation matrix of size $K \times K$, where each coefficient is computed over T temporal frames. Therefore, the time complexity of this operation is $\mathcal{O}(K^2T)$. The existing method uses a static threshold to filter out

magnitude vectors, which are low correlation with the respiratory signal. In contrast, our adaptive method introduces additional computations of the mean and standard deviation of correlation coefficients, followed by a non-linear thresholding function to improve robustness. The time complexity of these additional operations is $\mathcal{O}(K)$, and since typically $K \ll T$, their impact on the overall complexity is negligible. As such, both methods share the same time complexity of $\mathcal{O}(K^2T)$.

- Crossover Point Refinement:** The overall process is iterative in nature; however, since the number of iterations is typically small in practical applications and the number of iterations for both algorithms are comparable, we focus our analysis on the time complexity of a single iteration. In the existing method, the minimum adjacent crossover point interval is determined based on the set of existing crossover points. The algorithm subsequently traverses each pair of adjacent points to identify and eliminate invalid crossover points. Since each comparison and decision operation can be performed in constant time, the overall time complexity of this process is $\mathcal{O}(M)$, where M denotes the number of crossover points. Our proposed method similarly performs a traversal of the crossover points. It first computes both the minimum and maximum valid intervals based on the estimated RR , which is also constant-time. As with the existing algorithm, the subsequent step involves comparing and determining whether each crossover point is invalid, which maintains a time complexity of $\mathcal{O}(M)$. Therefore, both methods exhibit the same time complexity of $\mathcal{O}(M)$.

It is worth noting that feature point optical flow tracking is a shared component in both the existing and proposed methods. As its implementation is identical in both cases, it is excluded from the complexity comparison. In summary, the proposed method maintains the same theoretical time complexity as the existing lightweight method, while offering significant improvements in robustness and the accuracy of RR estimation.

4 Experiments

Table 1 presents the full names corresponding to the abbreviations of the different optical flow methods. And Table 2 presents the full names corresponding to the abbreviations of the different methods for extracting RR from respiratory signals. These abbreviations will help clarify the presentation of the paper.

Table 1 Full names corresponding to the abbreviations of the different optical flow methods

No.	Abbreviation	Full name
1	OF	Original optical flow method
2	FSSOF	Feature point selection strategy based optical flow method [1]
3	CGOF	Correlation-guided optical flow method [1]
4	TOF	Optical flow method combining FSS and CG [1]
5	IFSSOF	Our improved feature point selection strategy based optical flow method
6	ICGOF	Our improved correlation-guided optical flow method
7	IOF	Our optical flow method combining IFSS and ICG

Table 2 Full names corresponding to the abbreviations of the different methods for extracting RR from respiratory signals

No.	Abbreviation	Full name
1	PC	Peak counting method
2	CP	Crossover point method
3	FFT	Fast Fourier transform method
4	NFCP	Negative feedback crossover point method [1]
5	INFCP	Our improved negative feedback crossover point method

4.1 Dataset

The dataset used in this experiment is the Large-scale Bed-side Respiratory Dataset in Intensive Care (LBRD-IC), established by Tan et al. [1]. The LBRD-IC dataset consists of 10,100 videos from 17 different subjects, including 6 female and 11 male patients, with ages ranging from 28 to 76 years. Each subject's video recordings were collected in different hospital wards, captured using surveillance cameras from varying distances and angles. The original resolution of the videos is 960×576 pixels. Due to privacy concerns, the videos were cropped to a resolution of 200×100 pixels, retaining only the chest and abdominal regions. The videos in the LBRD-IC dataset are recorded at a frame rate of 25 fps, with each video lasting 10 s. Fig. 5 shows the typical scenarios in LBRD-IC dataset. The ground truth for RR in each video of the LBRD-IC dataset was measured by bedside ventilators and manually corrected under the guidance of respiratory clinicians.

The LBRD-IC dataset is designed for studying non-contact respiratory monitoring algorithms focusing on the chest and abdominal regions. This dataset exhibits several notable characteristics: 1) the video resolution is relatively low; 2) subjects are frequently partially occluded by medical staff during video recording; 3) subjects exhibit movements such as turning and sitting up; 4) as the videos are collected from long-term surveillance camera recordings, the lighting conditions in the recording environments vary significantly.

**Fig. 5** Typical scenarios in the LBRD-IC dataset, including various lighting conditions, recording distances, and camera angles

4.2 Evaluation criteria

To evaluate the performance of the improved RR extraction algorithm, three commonly used error metrics were selected: Mean Absolute Error (MAE), Mean Squared Error (MSE), and Mean Relative Error (MRE).

MAE calculates the average of the absolute differences between the predicted and actual values, providing an overall measure of predictive accuracy. Mathematically, MAE is defined as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |RR_i^{\text{predict}} - RR_i^{\text{label}}|, \quad (17)$$

where RR_i^{label} is the label, RR_i^{predict} is the predicted RR, and n is the number of samples.

MSE computes the average of the squared differences between the predicted and actual values. Due to the squaring operation, MSE is more sensitive to larger errors, making it particularly useful when penalizing extreme deviations. Mathematically, MSE is defined as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (RR_i^{\text{predict}} - RR_i^{\text{label}})^2. \quad (18)$$

MRE calculates the ratio of the prediction error to the actual value, providing a measure of relative error. Mathematically, MRE is defined as:

$$\text{MRE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{RR_i^{\text{predict}} - RR_i^{\text{label}}}{RR_i^{\text{label}}} \right|. \quad (19)$$

4.3 Comparisons with existing methods

The combination of the optical flow and crossover point method is the typical method for estimating RR from RGB videos [1]. We include this method in our comparative analysis and refer to it as OF-CP. Tan et al. [1] proposed an enhanced version of this method by incorporating a feature point selection strategy and a correlation-guided method to improve the quality of the extracted respiratory signals. Additionally, they introduced a negative feedback mechanism into the crossover point method to eliminate erroneous crossover points, thereby improving the accuracy of RR estimation. As the method preceding our improvements, we include this method in the comparative analysis and refer to it as TOF-NFCP. Furthermore, we consider two more state-of-the-art (SOTA) algorithms for comparison as follows:

Table 3 Comparison of different algorithms for extracting RR from RGB videos on the LBRD-IC dataset

Methods	MAE (BPM)	MSE (BPM ²)	MRE
OF-CP [1]	5.547	51.13	0.3584
TOF-NFCP [1]	4.464	36.75	0.2647
OF-M1D [44]	4.637	40.39	0.2978
AF-QG [45]	4.269	33.21	0.2501
IOF-INFCP	4.034	30.34	0.2481

The superior results are highlighted in bold

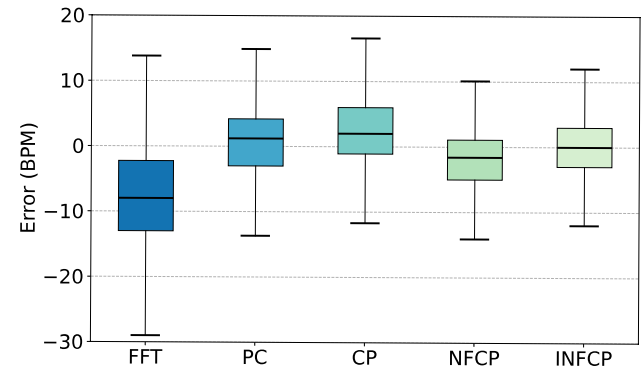


Fig. 6 Box plots of the error distribution of different RR calculation methods combined with TOF after removing outliers. The Error is defined as: Error = $RR_{\text{predict}} - RR_{\text{label}}$

Wang et al. [44] investigated motion-based respiratory signal extraction algorithms by evaluating six combinations derived from two motion estimation strategies and three spatial representations. Among these, the combination of optical flow and Multiple 1D profiles (M1D) demonstrated high sensitivity. M1D treats each image column as an independent vertical profile, so each patch contains multiple 1D profiles. Since the combination of optical flow and M1D has been shown to effectively eliminate motion artifacts, we include the method in the comparative study. In our study, we refer to this method as OF-M1D.

Song et al. [45] proposed a motion-robust video-based method for RR estimation by extracting and fusing respiratory signals from optical flow in the chest and abdomen regions, with motion artifacts suppressed via adaptive filtering and quality-guided signal selection. As this method shows superior performance and improves the practicality of video-based RR monitoring, we include it in our comparative analysis. In our study, we refer to this method as AF-QG. Since AF-QG relies on human keypoint detection to define ROIs, and the LBRD-IC dataset includes only the chest and abdomen regions with potentially severe occlusion, we use the entire region as the ROI when keypoint detection fails.

As shown in Table 3, the proposed IOF-INFCP algorithm achieves the best performance across all evaluation metrics. Specifically, it yields a MAE of 4.034 BPM, a MSE of 30.34 BPM², and a MRE of 0.2481, significantly outperforming

all compared methods. Notably, IOF-INFCP surpasses its predecessor TOF-NFCP by reducing the MAE from 4.464 to 4.034 and the MSE from 36.75 to 30.34, while also achieving a lower MRE. Moreover, it outperforms the current SOTA methods OF-MID and AF-QG, which previously demonstrated strong performance in RR estimation tasks. These results clearly highlight the enhanced accuracy and robustness of IOF-INFCP in estimating RR from RGB video on the LBRD-IC dataset. In the following sections, we further analyze the contributions of each methodological component, evaluate their individual impact on the overall performance, and examine how the proposed improvements affect the time complexity.

4.4 Key components analysis

4.4.1 Feature point selection strategy

To further validate the effectiveness of the dynamic tracking-enhanced feature point selection strategy, we combine the improved and original feature point selection strategies with different RR calculation methods and conduct experiments on the LBRD-IC dataset. As shown in Table 4, regardless of which RR calculation method is used, the improved

Table 4 Comparison of performance in RR estimation based on the LBRD-IC dataset using different feature point selection strategy optical flow methods and RR calculation methods

Methods	MAE (BPM)	MSE (BPM ²)	MRE
FSSOF-FTT	9.028	116.9	0.5309
FSSOF-PC	4.887	38.91	0.3100
FSSOF-CP	5.278	48.07	0.3423
FSSOF-NFCP	4.508	37.23	0.2676
FSSOF-INFCP	4.127	31.85	0.2534
IFSSOF-FTT	8.943	115.43	0.5237
IFSSOF-PC	4.798	37.33	0.3044
IFSSOF-CP	5.242	47.24	0.3400
IFSSOF-NFCP	4.376	35.39	0.2608
IFSSOF-INFCP	4.085	30.88	0.2534

The number of feature points is set to 5. The superior results are highlighted in bold

optical flow method consistently reduces the MAE, MSE, and MRE in RR estimation. This strongly demonstrates that the improved feature point selection strategy enhances the quality of feature points, thereby improving the quality of the respiratory signal.

As shown in Fig. 7, the feature points selected by the feature point selection strategy aggregating spatial information are often located along the edges of the chest or abdomen,

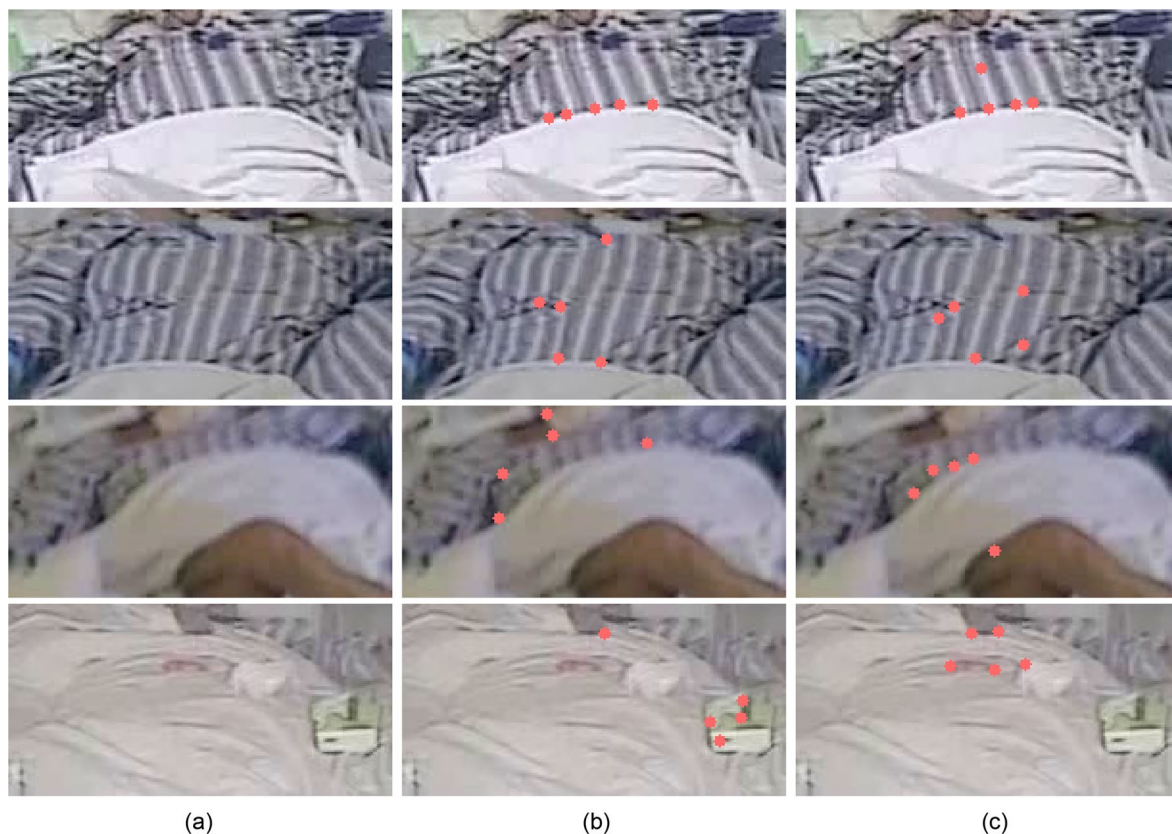


Fig. 7 Comparison of different feature point selection strategies. **a** First frame images of four different RGB videos. **b** Feature points selected by the feature point selection strategy aggregating spatial information. **c** Feature points selected by the improved feature point selection strategy

and sometimes even outside the body. In contrast, the feature points selected by our strategy are well-distributed on the chest or abdomen. This is because the corner response function used in the feature point selection strategy aggregating spatial information places too much emphasis on the texture features of corners, which weakens the guidance from spatial information, which causes the feature point quality function to favor points with significant texture variations. Our corner response function not only considers the texture characteristics of corners but also takes into account the dynamic tracking performance of these points. With the guidance of spatial information, our strategy selects feature points that are not only useful for optical flow tracking but also rich in respiratory information.

4.4.2 Correlation-guided methods

We visualized the magnitude vectors to compare the performance of two correlation-guided methods in optimizing feature point space. As shown in Fig. 8, compared to the correlation-guided method aggregating temporal information, our improved correlation-guided method demonstrates better consistency in the remaining magnitude vectors after removal, effectively discarding those with weak correlation to respiratory motion and those relatively more affected by noise. This results in a further improvement in the quality of the extracted respiratory signal. Additionally, to further assess the effectiveness of the adaptive correlation-guided method, we combined both the improved and

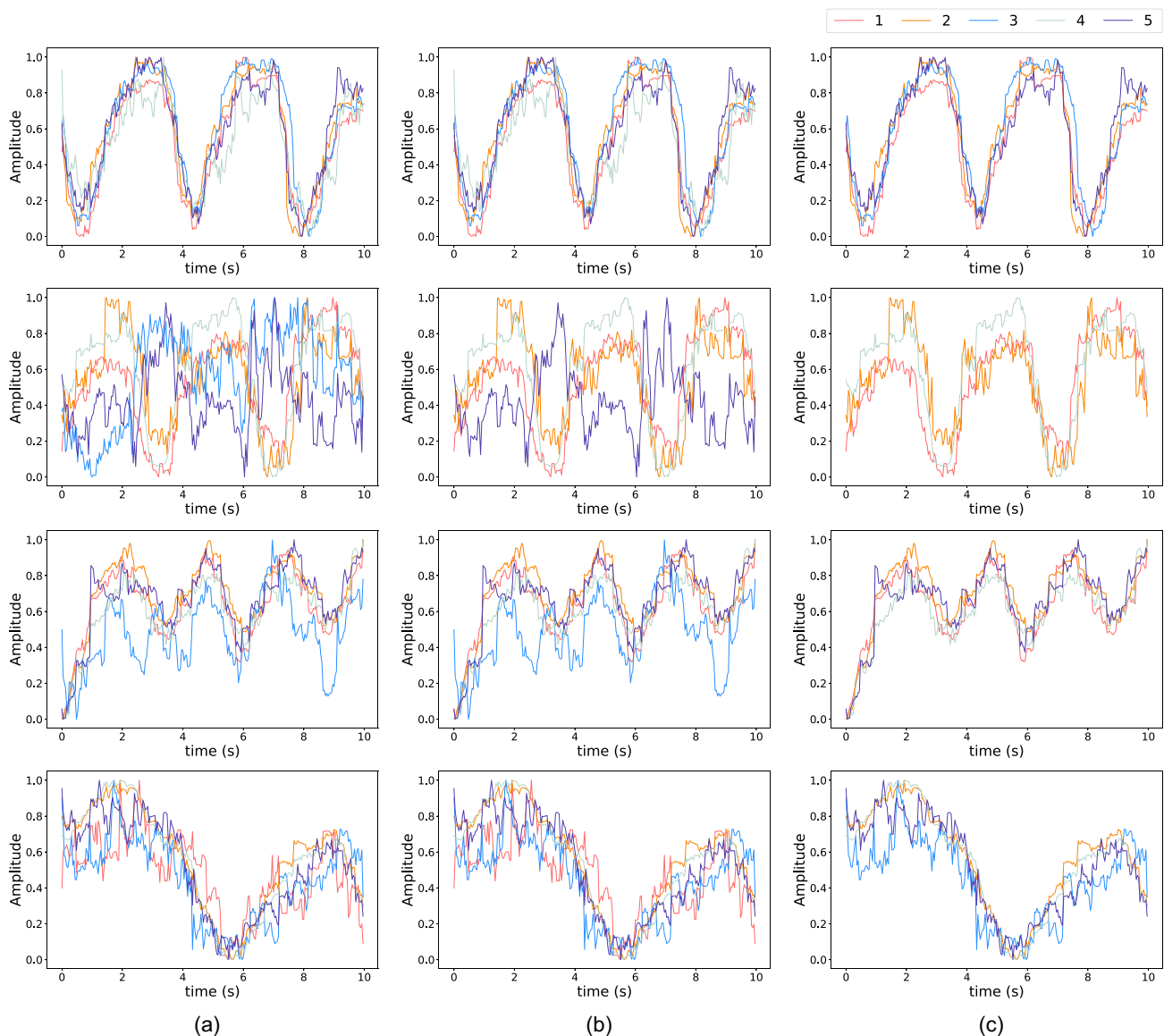


Fig. 8 Comparison of different correlation-guided methods when the number of feature points is 5. **a** Magnitude vectors corresponding to four RGB videos before feature point removal. **b** Magnitude vectors

remaining after processing with the correlation-guided method aggregating spatial information. **c** Magnitude vectors remaining after processing with the improved correlation-guided method

Table 5 Comparison of performance in RR estimation based on the LBRD-IC dataset using different correlation-guided optical flow methods and RR calculation methods

Methods	MAE (BPM)	MSE (BPM ²)	MRE
CGOF-FTT	8.968	115.9	0.5240
CGOF-PC	4.757	36.84	0.2983
CGOF-CP	5.048	43.73	0.3238
CGOF-NFCP	4.712	41.43	0.2768
CGOF-INFCP	4.229	33.12	0.2588
ICGOF-FTT	8.826	113.3	0.5145
ICGOF-PC	4.779	36.94	0.2973
ICGOF-CP	5.030	43.08	0.3208
ICGOF-NFCP	4.666	40.58	0.2748
ICGOF-INFCP	4.138	32.02	0.2514

The number of feature points is set to 5. The superior results are highlighted in bold

Table 6 Comparison of performance in RR estimation based on the LBRD-IC dataset using different optical flow methods and RR calculation methods

Methods	MAE (BPM)	MSE (BPM ²)	MRE
TOF-FTT	8.999	116.2	0.5287
TOF-PC	4.713	37.12	0.2972
TOF-CP	5.024	43.72	0.3241
TOF-NFCP	4.464	36.75	0.2647
TOF-INFCP	4.114	31.29	0.2531
IOF-FTT	8.922	114.4	0.5236
IOF-PC	4.733	36.39	0.2978
IOF-CP	4.996	42.83	0.3216
IOF-NFCP	4.421	36.49	0.2604
IOF-INFCP	4.034	30.34	0.2481

The number of feature points is set to 5. The superior results are highlighted in bold

original correlation-guided optical flow methods with different RR calculation methods and conducted experiments on the LBRD-IC dataset. As shown in Table 5, the improved correlation-guided optical flow method, when combined with most RR calculation methods, results in reductions in MAE, MSE, and MRE. This indicates that the improvement, achieved by dynamically adjusting the feature point

exclusion threshold, improves the accuracy of the synthesized respiratory signal.

4.4.3 Negative feedback crossover point methods

To evaluate the effectiveness of INFCP, we integrate TOF and IOF with various RR calculation methods. Regardless of the specific optical flow variant employed, INFCP consistently achieves the lowest error metrics across all RR calculation methods. In particular, the IOF-INFCP combination demonstrates the best performance, yielding a MAE of 4.034 BPM, a MSE of 30.34 BPM², and a MRE of 0.2481, which outperforms all other configurations. These results highlight a clear improvement over traditional CP and their predecessor, NFCP. Overall, the results confirm that INFCP enhances both the robustness and accuracy of RR estimation by incorporating physiological guidance and feedback, independent of the specific optical flow extraction technique used.

4.5 Influential analysis of feature point number

To explore the relationship between the number of feature points and the accuracy of the RR extraction methods, we varied the number of feature points and compared the changes in three evaluation metrics for both methods. As shown in Fig. 9, with the increase in the number of feature points, the estimation error of the proposed RR extraction method rapidly decreases. Moreover, when the number of feature points exceeds 5, the error stabilizes. In contrast, the original method shows that, as the number of feature points decreases, the error initially drops rapidly and reaches its minimum when the number of feature points is 5. However, as the number of feature points continues to increase, the error rises again. This can be attributed to the higher randomness and instability of both methods when the number of feature points is small, which leads to larger errors. As the number of feature points increases, the methods' stability

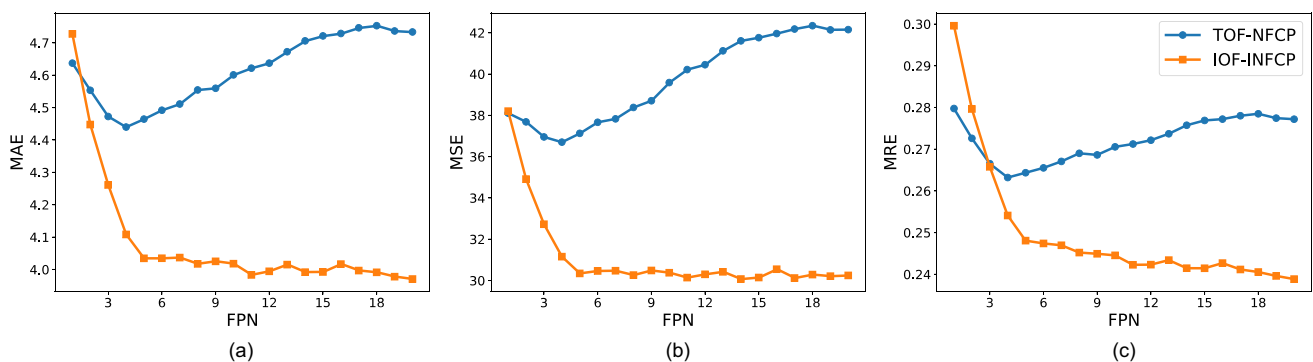


Fig. 9 The relationship between the number of feature points and RR extraction performance: **a** Mean absolute error (MAE), **b** Mean squared error (MSE), **c** Mean relative error (MRE) for the existing and improved methods. FPN represents the number of feature points

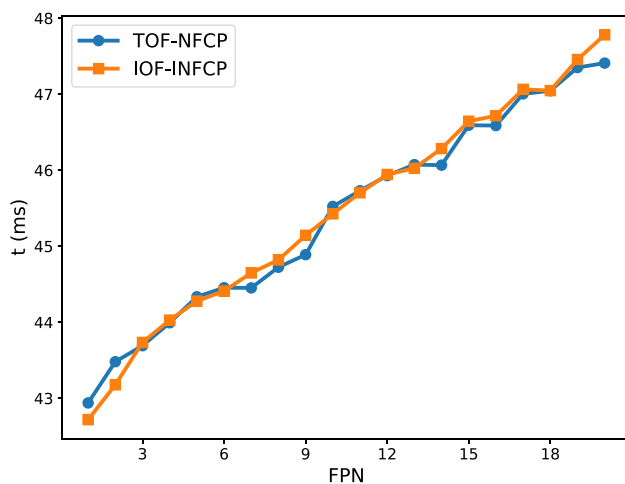


Fig. 10 Comparison of computing time between the existing TOF-NFCP method and our proposed IOF-INFCP method using different numbers of feature points

improves, resulting in a reduction in error. However, as the number of feature points increases, some points weakly correlated with respiratory motion and those heavily affected by noise are also included in the existing method. This leads to a degradation in the quality of the extracted respiratory signal and an increase in RR estimation errors. In contrast, the improved method mitigates the negative impact of increasing feature points on signal quality by selecting high-quality feature points. Additionally, by combining the adaptive correlation-guided method, we can dynamically adjust the threshold based on the overall signal quality, effectively removing feature points that are relatively weakly associated with respiratory motion or heavily affected by noise, thus further improving the quality of the respiratory signal. This also effectively demonstrates the validity of our dynamic tracking-enhanced feature point selection strategy and the adaptive correlation-guided method.

4.6 Influential analysis of computing time

To verify that the improvement strategy does not increase the algorithm's computational complexity, we conducted experiments on the LBRD-IC dataset, comparing the computing time of the two methods using different numbers of feature points. Specifically, for each selected number of feature points, we calculated the average time required to extract RR from the videos of LBRD-IC dataset. To minimize the influence of random fluctuations and ensure the reliability of our results, each experiment was repeated 10 times, and the mean values were reported. As shown in Fig. 10, the computing time of the two algorithms remain largely consistent regardless of the number of feature points. This indicates that our improved method enhances the accuracy of RR estimation without increasing the computational

cost, thereby validating the correctness of our theoretical analysis.

5 Conclusion

In this study, we propose the dynamic tracking-enhanced feature point selection strategy to better suit optical flow tracking. Additionally, we introduce the adaptive correlation-guided method to adapt to variations in different monitoring environments and individual differences. To address the issue of erroneous removal of valid crossover points in the negative feedback crossover point method, we utilize the physiological information of crossover points to identify erroneous crossover points, effectively preventing the removal of valid crossover points and thereby further reducing errors in RR calculation. Experimental validations on the LBRD-IC dataset show that the improved RR extraction algorithm significantly enhances both accuracy and robustness. The algorithm demonstrates higher precision in RR monitoring, thereby increasing the potential of non-contact RR monitoring in clinical applications and providing more reliable support for disease diagnosis and health monitoring.

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